

Mazes and counting

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In this guide to mathematics we will be exploring topics in counting by asking questions about mazes. As we go we will develop increasing algebraic sophistication, and touch base with other regions of mathematics. As a guide this can only scratch the surface of the sights available, it is up to the reader to visit personally for themselves, which is why a significant portion of the narrative is developed through exercises.

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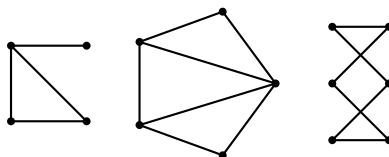
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1 Modeling mathematical objects

The material developed in the appendices lead to the ability to say quite a bit. I decided to start the paper here because this is where the fun begins, and everything that is necessary to understand before this is mostly just details and notation. With all of that out of the way we can develop a perspective of math through simple games.

1.1 Basic graph theory

When most people hear the word "graph" in they usually think about a box diagram, or perhaps a pie chart. Maybe they also think about the image of a curve, but today we will be discussing yet another kind of mathematical graph, used to represent relationships between objects. When we want to draw a graph, we usually draw the objects as vertices, and the relationships as edges.



More formally, a **graph** G is a set of vertices V and a set of sets E , each consisting of 2 vertices $\{u, v\}$, called the edge set. Since the edge set consists of ordered sets, we do not allow edges from a vertex to itself. If we are in a situation where we have could have a relationship from an object to itself we use multi sub sets of 2 vertices instead of sets. Normally we also do not think of having more than one edge between vertices but if we do want multiple edges, then we just let E be a multiset. The **degree** of a vertex $deg(v)$ is the number of edges containing that vertex as an element.

Exercise 1: First theorem of graph theory Prove that the sum of the degrees of vertices in a graph is equal to twice the number of edges:

$$\sum_{v \in V} deg(v) = 2|E|.$$

Exercise 2: Prove that in every graph there are at least 2 vertices with the same degree.

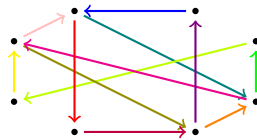
Two edges or two vertices in a graph are said to be **adjacent** if they share a vertex or edge respectively. The **neighborhood** $n(v)$ of a vertex $v \in V$ is the set of vertices adjacent to v . In our basic definition of a graph, $\deg(v) = |n(v)|$, but that is not always true for generalizations. A **path** in a graph is a nonempty sub ordered set of E such that consecutive edges are adjacent. If P is the path $\{\{v_0 v_1\}, \{v_1 v_2\}, \dots \{v_{n-1} v_n\}\}$, we call v_0 the start of the path and v_n the end of the path. A path is said to be **closed** when its start is the same as its end. Two vertices are said to be **path connected** if there is a path that contains both vertices.

Exercise 3: Prove that path connectedness is an equivalence relation on the vertices of the graph. These equivalence classes are known as the **connected components** or just components of the graph. A graph with a single component is called **connected**.

In most textbooks on graph theory, a path is not allowed to visit a vertex more than once, but here we will say in that case a path is **simple**. A path that is closed and simple is called a **cycle**. Similarly to exercise 3, we can define an equivalence relation **cycle connected** which holds whenever two vertices lie on a common cycle.

Exercise 4: Draw a graph with more cycle connected equivalence classes than components.

Exercise 5: A graph is called **Eularian** if there is a closed path which contains all of its edges, and such a path is called an **Eularian circuit**. Prove that a graph is Eularian if and only if it is connected and the degree of all of its vertices are even.



Exercise 6: A connected graph is called a **tree** if it contains no cycles. Prove that a tree with n vertices must have exactly $n - 1$ edges. Prove that for $n \geq 2$ if you have n distinguishable vertices, then there are $T_n = n^{n-2}$ possible trees that can be drawn using all n vertices.

Hint Pick two (not necessarily distinct) distinguished vertices to serve as roots of your tree, and count $n^2 T_n$.

Exercise 7 (hard): A graph is said to be **planar** if it can be drawn in the plane with no edges crossing. Let f be the number of regions enclosed by drawing the graph with no edge intersections, e be the number of edges in your graph, and v be the number of vertices. Prove that $f - e + v = 1$ for any connected planar graph.

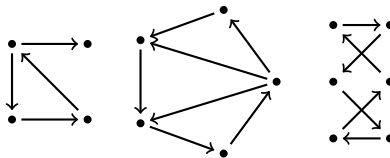
Hint Triangulate the graph, prove this preserves $f - e + v$, then find an algorithm which removes pieces of the perimeter but again preserves $f - e + v$. Observe $f - e + v = 1$ for a planar triangle.

Exercise 8 (harder): If we repeated exercise 7 but this time drawing our graph with no edge intersections on a sphere, we would observe $f - e + v = 2$ (why?). Conclude that there are exactly 5 platonic solids in three dimensions.

Hint Platonic solids are very symmetric, use that to make a Diophantine equation.

1.2 Digraphs and mazes

Sometimes we want the edges in our graph to have a front and a back. Given a graph $G = (V, E)$, then we have $2^{|E|}$ ways we can assign an orientation to each edge.



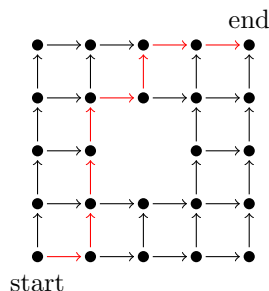
A **digraph** D is a set of vertices V and a set of lists E each consisting of 2 vertices $[u, v]$, called the edge set.

In the edge $e = [u, v]$, we say that v is the tip of the edge, and u is the tail, or that e points from u to v . For a vertex v we say that its in-neighborhood $inn(v)$ is the set of vertices that have an edge pointing at v , and $indeg(v)$ is the number of edges with v as their tip. Similarly the out-neighborhood $out(v)$ is the set of vertices being pointed at by an edge pointing from v , and $outdeg(v)$ is the number of edges with tail v .

Exercise 9: Reformulate the first theorem of graph theory for digraphs.

For a digraph we say that a path is sub ordered set of E such that the tip of any edge is the tail of the next edge, if a next edge exists. Otherwise the definition of a cycle is unchanged, but our idea of path connected needs to evolve if we dont want it to behave strangely. For instance, between two vertices u and v there may be a path starting at u and ending at v , but not the other way around unless they lie on a cycle. This means path connected is no longer an equivalence relation, though cycle connected still is. Therefore we say that the connected components of a digraph are the connected components of the graph you get by ignoring edge orientations.

If a digraph M contains no cycles and has a vertex u with indegree 0 as well as another vertex v with outdegree 0, then we can call M a **maze** with start u and end v . A **solution** of a maze is a path from start to end.



Usually it is not that hard to find a solution to a mathematical maze, the question we are interested in is how many solutions does a maze M have? Even for the simple example above, counting the number of solutions to a maze using brute force is usually an annoying task. If the start and end vertices are in different components of the digraph, then there are no solutions, so we can safely assume that all our mazes are connected, but that does not help us that much.

1.3 Water pipes

Fortunately, there is a simple algorithm that will always find the number of solutions, colloquially known as the water pipes algorithm. The process goes like this:

Define the weight of the start vertex $w(u)$ to be 1, and W to be set of all vertices without a weight whose in-neighborhoods consist of vertices which all have a defined weight. If there are any points besides the start with an empty in-neighborhood, weight them 0.

1. for each vertex x in W define the weight of x to be

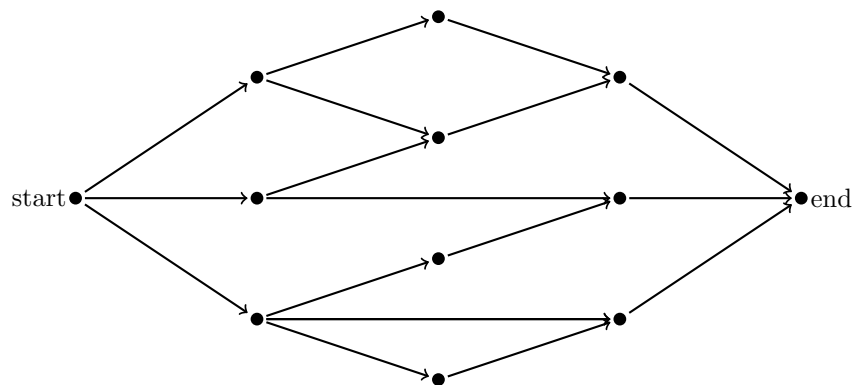
$$w(x) = \sum_{y \in \text{inn}(x)} w(y).$$

2. update W .

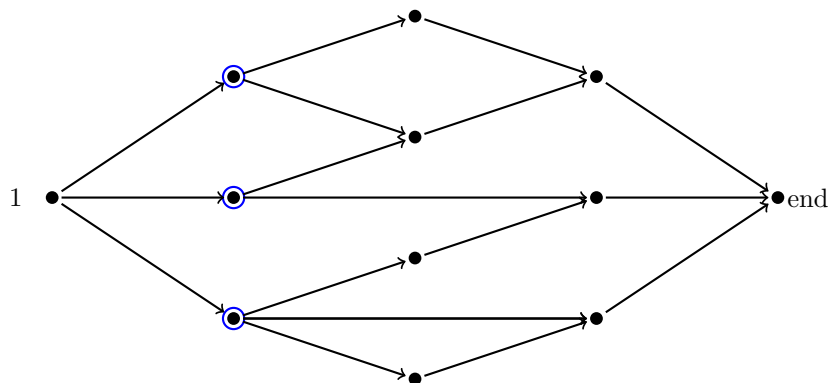
Repeat these steps until the end vertex v is in W , then the number of solutions to the maze is

$$w(v) = \sum_{y \in \text{inn}(v)} w(y).$$

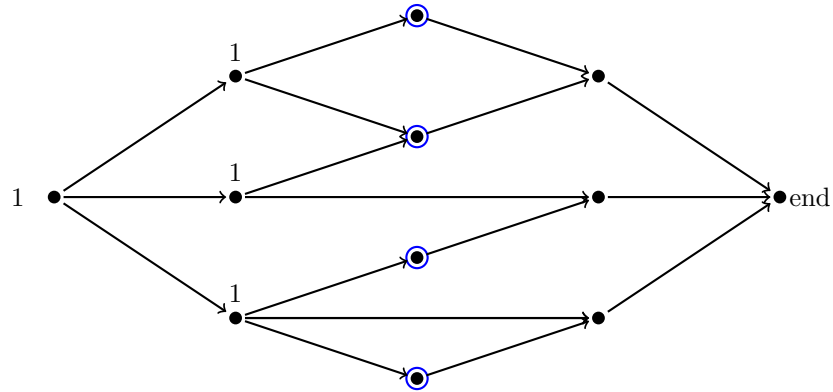
Lets see this in action on a simple example. Consider the following maze M :



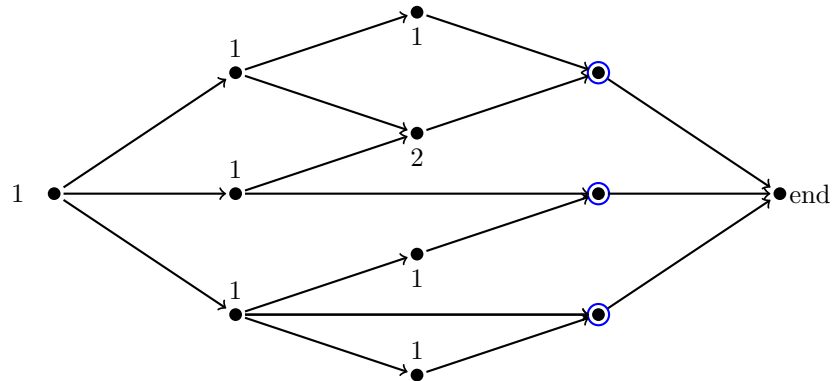
We start the water pipes algorithm by putting a 1 at the start and circling the elements of W .



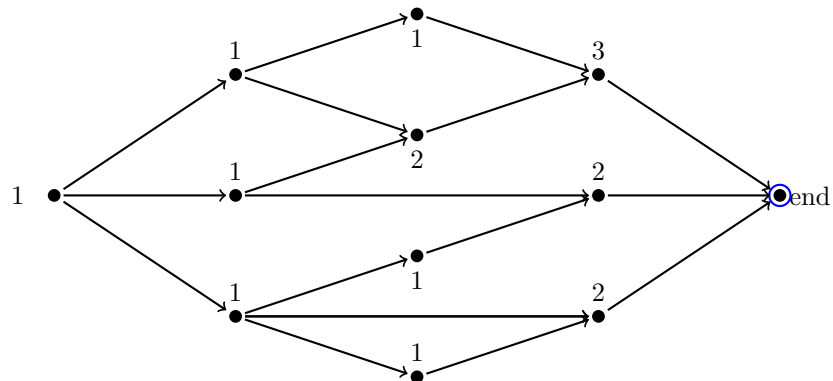
Each of these circled vertices gets a single 1 for their weights because their in-neighborhoods consist of just the start. We also circle the new elements of W ,



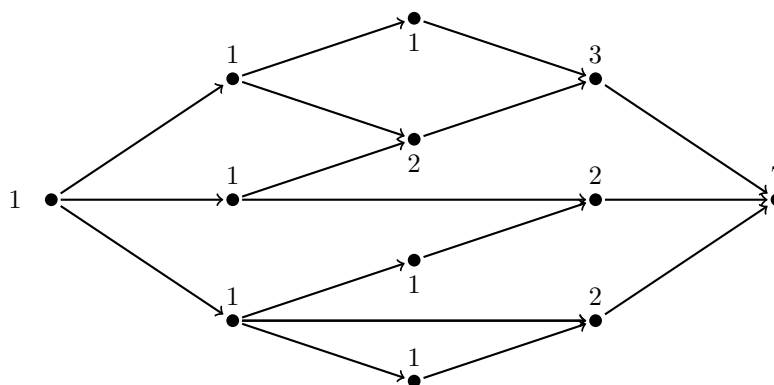
This time not every vertex gets weighted with 1



No vertex gets labeled with a 1 in this step



And finally we can read off that there are 7 solutions to this maze.

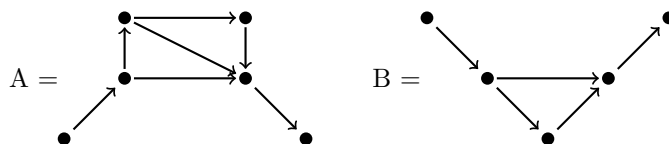


This algorithm is practical because it always solves the problem, and also creates a foundation to recast familiar counting questions in a more playful setting.

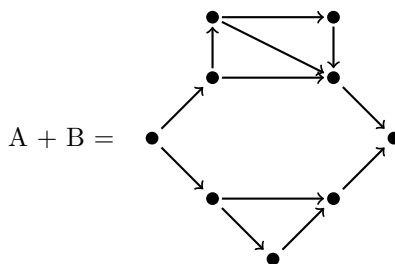
Exercise 10: Prove the water pipes algorithm given above always finds the number of solutions to a maze.

1.4 Maze composition

Before we conclude our discussion of mazes in general, it is worth it to briefly discuss ways of composing mazes and other exotic features. Suppose that the symbol $|M|$ is the number of solutions to a given maze, then for mazes A and B , define a new maze $A + B$ obtained by identifying both the start and end of A with both the start and end of B respectively. This is a "good" definition of the sum of mazes because we get $A + B = B + A$ and $|A + B| = |A| + |B|$. For example if



then we have $|A| = 3$, $|B| = 2$, and



which has $|A + B| = 5$.

There is one slight problem with this definition, and that happens when both A and B have an edge directly from start to end.

If we start with

$$A = B = \bullet \longrightarrow \bullet$$

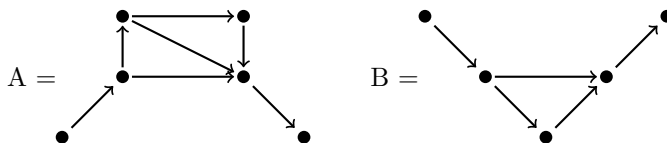
then by our definition we should have

$$A + B = \bullet \begin{array}{c} \circlearrowleft \\ \circlearrowright \end{array} \bullet$$

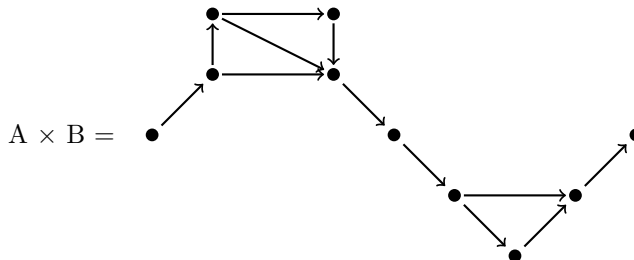
The fix is to change the definition of weight in the water pipes algorithm to accommodate multi digraphs, which is actually simple. Suppose $E(u, v)$ is the the multiset of directed edges from u to v in G , then all we need to do to modify the definition of weight is say

$$w(x) = \sum_{y \in \text{inn}(x)} |E(y, x)| w(y).$$

We can also define the product $A \times B$ of two mazes A and B to be the maze obtained by identifying the end of A with the start of B . With this definition we do not have $A \times B = B \times A$ for all mazes, but we do have $|A \times B| = |B \times A|$. Using the same mazes as before



we would have



and $|A \times B| = 6$.

Exercise 11: prove that $A \times C + B \times C$ is never equal to $(A + B) \times C$, but we always have $|A \times C + B \times C| = |(A + B) \times C|$ (using the usual order of operations).

Exercise 11 suggests that there may be a way to factor some mazes into 3 smaller mazes and recombine them into fifth smaller maze with the same number of solutions as the original. The contents of this section will be relevant later when we want the water pipes algorithm to apply to multi digraphs or mazes with restricted positions. For now we will close the section with two last applications of digraphs to relevant mathematical objects.

1.5 Modeling functions with digraphs

Given two sets $A = \{a_1 a_2 \dots a_n\}$ and $B = \{b_1 b_2 \dots b_k\}$, a **function** f from A to B is a subset of the Cartesian product

$$f = \{ (a_i, b_{j_i}) \}_{i=1}^n, \quad a_i \in A, b_{j_i} \in B$$

where the j_i 's range from 1 to k and do not have to be distinct. As shorthand we write $f(a_i) = b_{j_i}$, and if S is a subset of A , then the notation $f(S)$ is the set $\{f(s) \mid s \in S\}$. The set A is the **domain** of f , the set B is its **codomain**, and the set $I(f) = f(A) \subseteq B$ is the **image** of A determined by f in B . If the image of $f(a) = b$ is the domain of $g(b) = c$ we can define the composition $(g \circ f)(a)$ to be

$$(g \circ f)(a) = g(f(a)) = g(b) = c.$$

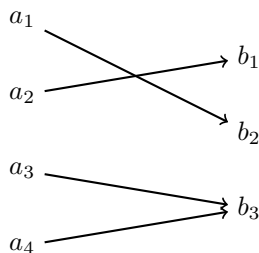
It is straightforward to show this definition is associative. We have a special function ι from X to itself, called the **identity function**, defined by $\iota(x) = x$ for any $x \in X$.

Exercise 12: Let F_K^N be the set of functions from N to K , where $|N| = n$ and $|K| = k$, then prove $|F_K^N| = k^n$.

One helpful way of visualizing a function f is using a digraph whose vertices are $V = \{a_1 \dots a_n b_1 \dots b_k\}$ and has a n directed edges, one for every list in f . Consider for example the function

$$f = \{ (a_1, b_2) (a_2, b_1) (a_3, b_3) (a_4, b_3) \}$$

where $A = \{a_1 a_2 a_3 a_4\}$ and $B = \{b_1 b_2 b_3\}$. Our digraph interpretation of it would be



A function f is said to be an **injection** when every element of B is pointed at by at most 1 arrow from A . If the size of A is bigger than the size of B , then there are 0 injections from A to B .

A function f is said to be a **surjection** when every element of B is pointed at by at least 1 arrow from A . If the size of A is less than the size of B , then there are 0 surjections from A to B . The first digraph we drew is a surjection.

A function f is said to be a **bijection** when every element of B is pointed at by exactly 1 arrow from A . If the size of A is not equal to the size of B , then there are 0 bijections from A to B . A function that is a bijection from a set to itself is called a **permutation**.

Exercise 13: Let I_K^N denote the set of injections from a the set N to the set K . What is $|I_K^N|$? How many permutations of a set of size n are there?

There are times we want to reverse the directed edges function, but the issue with that is with surjections we can get multiple arrows emerging from a single element. This means we cannot

always define an inverse function of $f(a) = b$, but we can define the **fiber** $f^{-1}(b)$ of b determined by f as

$$f^{-1}(b) = \{ a \in A \mid f(a) = b \}$$

We can go further and say in the equation $(g \circ f)(x) = x$ that g is a **right inverse** of f , as well as f is a **left inverse** of g .

Exercise 14: Prove that f has a left inverse if and only if it is an injection, and has a right inverse if and only if it is a surjection.

Since a permutation is both an injection and a surjection, it has both a left and right inverse.

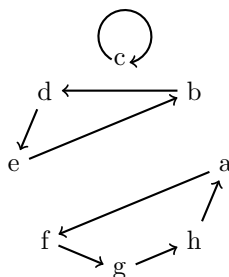
Exercise 15: Given that f is a permutation, g is a right inverse of f , and h is a left inverse, show that $g = h$ and that this function is unique.

A permutation σ of the set of letters $N = \{ a \ b \ \dots \ n \}$ can be associated to a size n ordered subset of N given by

$$\{ \sigma(a), \sigma(b), \dots, \sigma(n) \}$$

or just the word $\sigma(a)\sigma(b)\dots\sigma(n)$. Since there are only finitely many outcomes for a permutation to rearrange the elements in a set, and every permutation is invertible, eventually by composing a permutation with itself you must obtain the identity function. The **order** of a permutation $order(\sigma) = n$ is the smallest positive integer such that composing n copies of σ is the identity function.

We can associate another digraph to permutations which reveal additional insights about their structure. Given n vertices labeled with the elements of N , we define an edge for every ordered pair $(k, \sigma(k))$. So for example, consider the permutation sending $abcdefgh$ to $fdcebgha$. Our construction gives us the corresponding digraph



We can see that the effect of a permutation is move the letters in cycles (we count the fixed points as their own cycle). This suggests another way of writing of writing a permutation according to its cycles. In this case we would have $(afgh)(bde)(c)$ which represents each cycle in the digraph with paranthesis with the elements in a cyclic order such that

$$(i_1 i_2 \dots i_k) = (i_k i_1 \dots i_{k-1}).$$

This makes the representation unique up to the chosen internal and external ordering of the cycles. This representation is also called a **cycle partition** of n things into k cycles because each cyclic subset of N is disjoint from the rest and their union would contain everything in N .

1.6 Sign reversing involutions

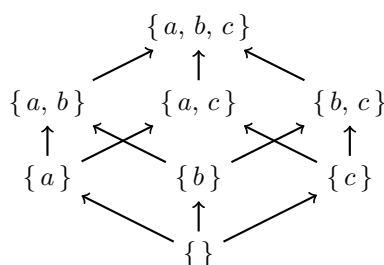
1.7 Partial orders and sums of divisors

As interesting as equivalence relations and functions are there are more general types of relationships between sets we want to model. We will also see how digraphs can help us understand theorems in number theory. Familiarity with series is recommended, which are covered in appendix c. A partial order $\leq$ on a set X is a relation on $X \times X$ that satisfies the following 3 properties:

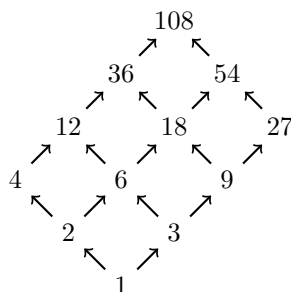
1. **Reflexivity:** $a \leq a$
2. **Anti-symmetry:** If $a \leq b$ and $b \leq a$ then $a = b$
3. **Transitivity:** If $a \leq b$ and $b \leq c$ then $a \leq c$.

as suggested by our overloaded notation, less than or equal to is the prototypical example of a partial order the same way equals is the prototypical equivalence relation. Two more important examples of partial orders are the subset relation $A \subseteq B$ and the divides relation $k|n$ from number theory. A set with a partial order is called a partially ordered set, or a poset.

We can draw a digraph representing the structure of a partial order called a **Hasse diagram**. If we have a partial order $\leq$ on a set $X = \{1 2 \dots x\}$, we draw a digraph whose vertices are labeled with elements of X and has an edge between every two distinct vertices u and v whenever $u \leq v$ and there does not exist a w distinct from u and v such that $u \leq w \leq v$. For example, suppose that X is the power set of $\{a b c\}$ and our partial order is the subset relation, then our Hasse diagram is:



It is not an accident that this Hasse diagram looks like a cube. If we draw the Hasse diagram of a set with n elements, it can always be drawn to look like a projection of the n -hypercube onto the plane. We can do this because we can associate every subset of $N = \{1 2 \dots n\}$ to elements to a list of size n that has a 0 in its k^{th} spot if k is not in the subset, and a 1 otherwise. If instead we chose our set to be the divisors of the number 108 and our partial order to be the divides relation, then our Hasse diagram is called a factor lattice and it looks like:



This picture is very helpful for helping us prove algebraic formulas about the divisors of a number.

The fundamental theorem of arithmetic tells us that any natural number larger than 1 can be written as a product of primes uniquely. Suppose we have that

$$n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$$

where the p_i 's are distinct primes, and $e_i \geq 1$, then we define the number of divisors function

$$\tau_0(n) = \sum_{d|n} 1.$$

The definition we gave is quite tedious to use because it requires us to have the whole set of divisors written down beforehand to use. Instead we can use the Hasse diagram. Thinking of the digraph as a k dimensional rectangular solid, we can see that it contains $(1 + e_1)(1 + e_2) \dots (1 + e_k)$ vertices, since we can think of a divisor $d = p_1^{f_1} \dots p_k^{f_k}$ where $0 \leq f_i \leq e_i$ as an integer point $(f_1, \dots, f_k)$. Therefore we have that

$$\tau_0 = (1 + e_1)(1 + e_2) \dots (1 + e_k)$$

We can also define a sum of divisors function

$$\tau_1(n) = \sum_{d|n} d$$

but this suffers from the same problem as the last definition. Looking back at the hasse diagram for the divisors of 108, we can see that every row going up and to the right contains increasing powers of 3 multiplied by a power of 2. When we add up these rows going to the right, lets parenthesize them together

$$\begin{aligned} \tau_1(n) &= (1 + 3 + 9 + 27) + (2 + 6 + 18 + 54) + (4 + 12 + 36 + 108) \\ &= (1 + 3 + 9 + 27) + 2(1 + 3 + 9 + 27) + 4(1 + 3 + 9 + 27) \\ &= (1 + 2 + 4)(1 + 3 + 9 + 27) = 7 \cdot 40 = 280. \end{aligned}$$

In general we can prove

$$\tau_1(n) = \prod_{i=1}^k (1 + p_i + \dots + p_i^{e_i}) = \prod_{i=1}^k \frac{p_i^{e_i+1} - 1}{p_i - 1}$$

by induction on the number of prime factors of n , but suggestions for how to proceed are given by the hasse diagram of the divisors of n .

Exercise 17: Define the sum of the j^{th} powers of the divisors of n by $\tau_j(n) = \sum_{d|n} d^j$. Show that we can rewrite this sum into the product $\tau_j(n) = \prod_{i=1}^k \frac{p_i^{j(e_i+1)} - 1}{p_i^j - 1}$.

These kinds of ideas can be generalized to a type of generating function called a **Dirichlet series**

$$A(s) = \sum_{n \geq 1} \frac{a_n}{n^s}.$$

This topic are a little more advanced a may not be suitable for a first read through the notes, though they are still worth considering in this context.

A function is said to be **number theoretic** if it maps natural numbers to natural numbers. A type of function $f(n)$ which is sensitive to the factor lattice of a natural number is **multiplicative** if $f(n) = f(p)f(q)$ whenever $n = pq$ and $\gcd(p, q) = 1$. The sum of the k^{th} powers of the divisors of a natural number is an example of a multiplicative number theoretic function, and so is the function which is uniformly 1. Multiplicative number theoretic functions $f(n)$ are completely determined by how they act on powers of primes, and so we get the following formal series identity

$$\sum_{n \geq 1} \frac{f(n)}{n^s} = \prod_{p \geq 1} (1 + f(p)p^{-s} + f(p^2)p^{-2s} + \dots).$$

Exercise 18 Prove the previous two sentences.

Being so familiar with enumerating factors of a number comes in handy because we get a very general convolution identity

2 Subset recurrence relations

2.1 Basic types of subset structures

In these subsections we will be concerned with 3 basic ways of splitting up the set

$$N = \{1\ 2\ \dots\ n\}$$

of n distinguishable objects into specialized subsets. Define the following symbols to mean

1. $\binom{n}{k}$ = the number of ways pick a subset of size k from a set of size n .
2. $\{n\}_k$ = the number of ways to pick k disjoint nonempty subsets from a set of size n .
3. $[n]_k$ = the number of ways to pick k disjoint nonempty cyclic subsets from a set of size n .

Respectively these symbols are called the binomial coefficients and the factorial coefficients of the second and first kinds, for reasons that are not important now. The symbols are read as "n choose k", "n set partition k", and "n cycle partition k". The names Blaise Pascal, Omar Khayyam, Yang Hui, Acharya Pingala, James Stirling, and Ivo Lah should also be mentioned.

For all of these coefficients it is helpful to arrange them in a descending triangle

$$\begin{array}{cccccccccccccccc}
 & & & & \binom{0}{0} & & & & \{0\}_0 & & & & [0]_0 & & & & \\
 & & & \binom{1}{0} & \binom{1}{1} & & & & \{1\}_0 & \{1\}_1 & & & [1]_0 & [1]_1 & & & \\
 & & \binom{2}{0} & \binom{2}{1} & \binom{2}{2} & & & & \{2\}_0 & \{2\}_1 & \{2\}_2 & & [2]_0 & [2]_1 & [2]_2 & & \\
 \binom{3}{0} & \binom{3}{1} & \binom{3}{2} & \binom{3}{3} & & & \{3\}_0 & \{3\}_1 & \{3\}_2 & \{3\}_3 & & & [3]_0 & [3]_1 & [3]_2 & [3]_3 &
 \end{array}$$

which is also a visualization technique we will employ for more complicated counting triangles.

Exercise 18: Find all the values in the first four rows of the triangles above.

Later on we will also find reason to discuss a fourth type of counting symbol;

4. $L(n, k)$ = the number of ways to pick k disjoint nonempty sub orderedsets from a set of size n .

but we will also need some new notation for factorials. For most applications, the exclamation mark taken to mean

$$n! = (n)(n-1)(n-2)\dots(2)(1), \ 0! = 1$$

for any natural number n , works well enough to make additional sophistication burdensome. However, we are frequently interested in expressions like

$$n(n-1)(n-2)\dots(n-k+1) = \frac{n!}{k!}$$

but since that expression is an integer, we would prefer to have a way for writing it which does not use a fraction bar. To that end we introduce the notations for rising and falling factorials starting at n with k terms.

$$\begin{aligned}
 n^{\overline{k}} &= n(n+1)(n+2)\dots(n+k-1), \\
 n^{\underline{k}} &= n(n-1)(n-2)\dots(n-k+1).
 \end{aligned}$$

Notice that $n^{\underline{k}}$ counts the number of ways to pick k objects from n in some order without replacement, as well the number of injections from n objects to k .

2.2 The binomial coefficients

The purpose of the previous exercise is not to be excessively cruel, but to convince you that we should find a smart way to compute these numbers.

Exercise 19: Prove $\binom{n}{k} = \frac{n^k}{k!}$. This computational formula lets us generalize the binomial coefficients to complex values of n .

Exercise 20: For n a natural number, prove that $\binom{-n}{k} = (-1)^k \binom{n+k-1}{k-1}$.

Exercise 21 (hard): Prove that $\binom{\frac{1}{2}}{k} = \frac{(-1)^{k+1}}{4^k(2k-1)} \binom{2k}{k}$.

Having a formula is nice and all, but sometimes it is not as helpful as we would like. In order to avoid two big multiplications and a division we would need to cancel common factors from the numerator and denominator. While that by itself is not necessarily a painless task, on top of that we have the problem that without prior knowledge of this number counting the cardinality of a set, we have no guarantee at this time that all the factors will cancel. To this end we will develop a recurrence relation that uses previously known values to compute new ones.

Lets get a couple of basic ideas straight. Since deciding to take k things is the same as deciding to leave behind $n - k$ things, we have $\binom{n}{k} = \binom{n}{n-k}$, and since there is only one way each to take or leave it all, we have $\binom{n}{0} = \binom{n}{n} = 1$. With these considerations in mind we can deduce the following

Lemma 1: The binomial coefficients satisfy the following recurrence relation:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Proof A common strategy for proving arithmetical statements involving counting symbols is to ask "What is this counting?", assuming you are given the identity ahead of time. Here we see that the left side is counting the number of ways to pick a subset of k objects out of n . Lets see if we can come up with another way to count these subsets. Lets write our set of n elements as

$$N = M \cup \{n\}.$$

where $M = \{1 2 \dots n-1\}$. The subsets we are interested can be split into two cases, depending on whether they contain n or not.

If a subset $S \subseteq N$ contains n , then the rest of its $k-1$ elements must be chosen from M , which there are $\binom{n-1}{k-1}$ ways to do so. If the subset S does not contain n , then all k elements must come from M and there are $\binom{n-1}{k}$ to do that. Summing over the cases achieves the result.

This strategy is often referred to as a "counting argument", and it forms the backbones of many results that are widely applied outside of discrete math.

Exercise 22: Prove that $\binom{n}{k} = \sum_{j=0}^i \binom{i}{j} \binom{n-i}{k-i+j}$ for $i \geq 1$.

In the next exercise we will see another type of method of proving counting statements.

Exercise 23 (Subset pgf): A **polynomial generating function** (or **pgf** for short) is a polynomial whose terms encode casework for a counting problem. Prove the following identity

$$\prod_{i=1}^n (1 + x_i) = \sum_{S \subseteq N} \prod_{j \in S} x_j$$

and conclude $(1 + x)^n = \sum_{i=0}^n \binom{n}{i} x^i$.

Exercise 24 (Binomial theorem): Use a counting argument to prove

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}.$$

Hint Consider lists of size n whose elements come from two sets $|A| = a$ and $|B| = b$.

Exercise 25 (Factorial theorem): Use a counting argument to prove

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}.$$

Hint Consider ordered sets of size n whose elements come from two sets $|A| = a$ and $|B| = b$.

We are interested in these next two identities later, here we will ask the reader to prove them twice using first a counting argument, then a pgf. We will look at yet another proof of both later.

Exercise 26 (Vandermonde convolution): Prove the following in two different ways

$$\binom{m+n}{k} = \sum_{i=0}^k \binom{m}{i} \binom{n}{k-i}.$$

Hint Consider $(1 + x)^{m+n} = (1 + x)^m (1 + x)^n$.

Exercise 27 (Hockey stick identity): Prove that

$$\binom{n+1}{k+1} = \sum_{i=k}^n \binom{i}{k}.$$

Hint Consider

$$X^k + X^{k+1} + \dots + X^n = \frac{X^k - X^{n+1}}{1 - X}$$

and let $X = 1 + x$. What is the coefficient of x^k ?

Exercise 28 (hard): Using the previous exercise, prove that we can sum over any rectangle in pascals triangle just by knowing values near the corners

$$\sum_{i=r}^m \sum_{j=s}^n \binom{i+j}{i} = \binom{m+n+2}{n+1} - \binom{r+n+1}{r} - \binom{m+s+1}{s} + \binom{r+s}{r}.$$

Compare with the fundamental theorem of calculus.

2.3 The factorial coefficients of the second kind

With that review of the binomial coefficients concluded, we can briefly discuss the factorial coefficients.

At the beginning of the section we introduced $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ as the number of ways to split n distinguishable objects into k nonempty disjoint subsets known as **blocks**. Such a configuration is called a **set partition** from n of size k . For historical reasons these are referred to as factorial coefficients of the second kind, but we will see why later. Right now we want to mimic the thinking we did with the binomial coefficients to find a way to create new values out of old ones.

In this situation we would have $\left\{ \begin{smallmatrix} n \\ n \end{smallmatrix} \right\} = 1$ for $n \geq 0$, the argument being would be that there is 1 way to put everything in a set into a subset by itself, with the agreement that there is exactly one way to put everything in the empty set into a set by itself. However, this time we would also get that $\left\{ \begin{smallmatrix} n \\ 0 \end{smallmatrix} \right\} = 0$ since for $n > 0$ clearly there are no ways to have zero subsets with a nonzero amount of stuff in them. So this time we would not expect our triangle to be symmetric.

Exercise 29: Use counting arguments to prove:

1. $\left\{ \begin{smallmatrix} n \\ 1 \end{smallmatrix} \right\} = 1$
2. $\left\{ \begin{smallmatrix} n \\ 2 \end{smallmatrix} \right\} = 2^{n-1} - 1$
3. $\left\{ \begin{smallmatrix} n \\ n-1 \end{smallmatrix} \right\} = \binom{n}{2}$

Lets try to find an explicit formula similar to our first binomial exercise. We will do this by reinterpreting what we can count with $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$.

Exercise 30: Let E_K^N be the set of equivalence relations on n objects with k equivalence classes, prove $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} = |E_K^N|$. Conclude that $k! \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ counts the number of surjections from an n element set to a k element set.

In order to come up with a formula for $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ we will count the surjections from a set with n elements to a set using the principle of inclusion/exclusion, which is reviewed in the appendix of counting principles.

Let F_K^N be the set of all functions from the set $N = \{1 2 \dots n\}$ to the set $K = \{1 2 \dots k\}$, so that $|F_K^N| = k^n$. Define the set F_i^n to be the set of all functions f from N to K such that i is not mapped to by f . Then the number we are looking for is $k^n - \left| \bigcup_{i=1}^k F_i^n \right|$ because we need every element of K to be mapped to. According to the principle of inclusion/exclusion

$$\left| \bigcup_{i=1}^k F_i^n \right| = \sum_{\substack{J \subseteq K \\ J \neq \{\}}} (-1)^{1+|J|} \left| \bigcap_{j \in J} F_j^n \right|.$$

We can rewrite this sum by looking at the size of J . The size of the intersection of the F_j 's are the number of functions from N to the set $K - J$, of which there are $(k - |J|)^n$. If we let $|J| = i$ then there are $\binom{n}{i}$ ways to pick J , so casework tells us

$$\sum_{\substack{J \subseteq K \\ J \neq \{\}}} (-1)^{1+|J|} \left| \bigcap_{j \in J} F_j^n \right| = \sum_{i=1}^k (-1)^{1+i} \binom{n}{i} (k-i)^n$$

and we can conclude

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{1}{k!} \left[k^n - \sum_{i=1}^k (-1)^{1+i} \binom{n}{i} (k-i)^n \right] = \frac{1}{k!} \sum_{i=0}^k (-1)^i \binom{n}{i} (k-i)^n.$$

Like last time, this formula can be more unwieldy than is required for our purposes, so we will basically forget about it for now.

Exercise 31: Use a counting argument to prove

$$x^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k.$$

What we really want is a recurrence relation letting us generate new values out of old values.

Lemma 2 The factorial numbers of the second kind satisfy the following recurrence relation:

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \left\{ \begin{matrix} n-1 \\ k-1 \end{matrix} \right\} + k \left\{ \begin{matrix} n-1 \\ k \end{matrix} \right\}.$$

Proof We will use the same strategy as lemma 1. Notice that the left side is counting the number of ways to pick a set partition of n objects into k blocks. Lets see if we can come up with another way to count these set partitions. Lets write our set of n elements as

$$N = M \cup \{n\}.$$

where $M = \{1, 2, \dots, n-1\}$. The set partitions we are interested can be split into two cases, depending on whether n is in a block by itself or not. If it is by itself, then there are $\left\{ \begin{matrix} n-1 \\ k-1 \end{matrix} \right\}$ ways to pick the remaining blocks from the elements of M . If n is not by itself then it has to be added to an already existing block. First we have to pick k blocks from the $n-1$ objects of M , then select which block we are inserting n into. There are $k \left\{ \begin{matrix} n-1 \\ k \end{matrix} \right\}$ ways of doing so, completing the proof.

Exercise 32: Use a counting argument to prove

$$\left\{ \begin{matrix} n+1 \\ k+1 \end{matrix} \right\} = \sum_{j=k}^n \binom{n}{j} \left\{ \begin{matrix} j \\ k \end{matrix} \right\}.$$

Exercise 33: Use a counting argument to prove

$$\left\{ \begin{matrix} n+1 \\ k+1 \end{matrix} \right\} = \sum_{j=k}^n (k+1)^{n-j} \left\{ \begin{matrix} j \\ k \end{matrix} \right\}.$$

Exercise 34: Use a counting argument to prove

$$\left\{ \begin{matrix} n+k+1 \\ k \end{matrix} \right\} = \sum_{j=0}^k j \left\{ \begin{matrix} n+j \\ k \end{matrix} \right\}.$$

2.4 The factorial coefficients of the first kind

The third counting symbol we want to discuss is the factorial number of the first kind. Recall that $\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]$ counts the number of ways to place n objects into k disjoint cyclic subset. Such a configuration is called a **cycle partition** on n with k cycles. As with permutations, we write cyclic subsets with parenthesis $(123) = (312) = (231) \neq (132)$. Like before we could find an explicit formula for $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$, but it would not be so useful to us. We will obtain an explicit closed form later when we find a pgf for these numbers. In this situation we again have that $\left[\begin{smallmatrix} n \\ n \end{smallmatrix} \right] = 1$ for any $n \geq 0$, and $\left\{ \begin{smallmatrix} n \\ 0 \end{smallmatrix} \right\} = 0$ for $n > 0$, which means we do not get symmetry again.

Exercise 35: Use counting arguments to prove:

1. $\left[\begin{smallmatrix} n \\ 1 \end{smallmatrix} \right] = (n-1)!$
2. $\left[\begin{smallmatrix} n \\ 2 \end{smallmatrix} \right] = (n-1)! \left(1 + \frac{1}{2} + \dots + \frac{1}{n-1} \right)$
3. $\left[\begin{smallmatrix} n \\ n-1 \end{smallmatrix} \right] = \binom{n}{2}$
4. $\left[\begin{smallmatrix} n \\ n-2 \end{smallmatrix} \right] = \frac{3n-1}{4} \binom{n}{3}$
5. $\sum_{k=0}^n \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] = n!$

Even without a formula we can be confident that we can compute these numbers because we will once again use a distinguished element to create a recurrence relation.

Lemma 3 The factorial numbers of the first kind satisfy the following recurrence relation:

$$\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] = \left[\begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix} \right] + (n-1) \left[\begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right].$$

Proof We will use the same strategy as lemma 1. Notice that the left side is counting the number of ways to pick a cycle partition of n objects into k cycles. Lets see if we can come up with another way to count these cycle partitions. The cycle partitions we are interested can be split into two cases, depending on whether n is in a cycle by itself or not. If it is by itself, then there are $\left[\begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix} \right]$ ways to pick the remaining cycles. If n is not by itself then it has to be added to an already existing cycle. First we have to pick k cycles from $n-1$ objects, then select which element we are inserting n in front of in its cycle. There are $(n-1) \left[\begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right]$ ways of doing so.

The next exercises will help us get our formula, and will also explain the ordering of the names of the factorial numbers.

Exercise 36: Use a counting argument to prove

$$x^{\overline{n}} = \sum_{k=0}^n \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] x^k$$

Hint The symbol $x^{\overline{n}}$ counts the number of ways n labeled flags can be placed on x labelled flagpoles.

Exercise 37: By setting x to $-x$ in previous exercises, prove that $x^{\underline{n}} = \sum_{k=0}^n (-1)^{n-k} \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] x^k$ as well as $x^{\underline{n}} = \sum_{k=0}^n (-1)^{n-k} \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} x^{\overline{k}}$. Prove these identities again with a sign reversing involution.

The reason that these are called the factorial numbers of the first kind because they take us from polynomials written as powers of x to polynomials written in terms of rising/falling factorials, and the second kind take us backwards. In James Stirling's first writing on this topic, he introduced these contexts in the same order we did, albeit without the same nice notation.

We are in a position to find a formula for $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ since it is the coefficient of x^k in $x^{\overline{n}}$. Recall from Vieta that

$$\prod_{i=1}^n (x - r_i) = \sum_{k=0}^n (-1)^{n-k} \left[\sum_{\substack{J \subseteq N \\ |J|=n-k}} \prod_{j \in J} r_j \right] x^k$$

where $N = \{1, 2, \dots, n\}$. By letting $r_i = 1 - i$ and comparing coefficients with $x^{\overline{n}}$ we get the formula

$$\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] = (-1)^{n-k} \sum_{\substack{J \subseteq N \\ |J|=n-k}} \prod_{j \in J} (1 - j).$$

Exercise 38: Use a counting argument to prove

$$\left[\begin{smallmatrix} n+1 \\ k+1 \end{smallmatrix} \right] = \sum_{i=k}^n \left[\begin{smallmatrix} n \\ i \end{smallmatrix} \right] \binom{i}{k}.$$

Exercise 39: Use a counting argument to prove

$$\left[\begin{smallmatrix} n+1 \\ k+1 \end{smallmatrix} \right] = \sum_{i=k}^n \left[\begin{smallmatrix} i \\ k \end{smallmatrix} \right] n^{\overline{n-i}}.$$

Exercise 40: Use a counting argument to prove

$$\left[\begin{smallmatrix} n+k+1 \\ k \end{smallmatrix} \right] = \sum_{i=0}^k (n+i) \left[\begin{smallmatrix} n+i \\ i \end{smallmatrix} \right]$$

Exercise 41: Explain why $\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right] \geq \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ for any $n, k \geq 0$. **Hint** think about cycle connected versus path connected.

Before we conclude this section and return to mazes we will discuss factorial numbers of the third kind, and take a first look at sum of powers formulas.

2.5 The factorial coefficients of the third kind

At the beginning of the section we introduced the symbol $L(n, k)$ to count the number of ways to split n distinguishable objects into k disjoint linearly ordered subsets. We call this kind of configuration a **line partition** of n with k lines. There are two different formulas for these numbers that we can use that have significant combinatorial interpretations, but we will only prove one and leave the other as an exercise.

As with the other factorial numbers we would expect that $L(n, n) = 1$ for any $n \geq 0$, but $L(n, 0) = 0$ for any $n > 0$. Now we can skip the process of finding specific formulas for specific

values of k . In order to partition n things into k lines, first arrange them into a single long line in one of $n!$ ways. Then we choose $k - 1$ spots to cut the line into k nonempty pieces. There are $\binom{n-1}{k-1}$ ways of doing this. Since the original order the lines appeared in does not matter, we have to divide by the $k!$ ways of ordering them. So we get

$$L(n, k) = \frac{n!}{k!} \binom{n-1}{k-1} = \binom{n-1}{k-1} n^{\underline{n-k}}.$$

Exercise 42: Use a counting argument and a pgf to prove following in two different ways

$$L(n, k) = \sum_{j=k}^n \begin{bmatrix} n \\ j \end{bmatrix} \left\{ \begin{matrix} j \\ k \end{matrix} \right\}.$$

These formulas are both intriguing, but once again we can also come up with a very simple recurrence relation that lets us generate these numbers without resorting to division or looking up values in tables we already have.

Lemma 4 The factorial numbers of the third kind satisfy the following recurrence relation:

$$L(n, k) = L(n-1, k-1) + (n+k-1)L(n-1, k).$$

Proof Notice that the left side is counting the number of ways to pick a line partition of n objects into k line. Lets see if we can come up with another way to count these line partitions. The cycle partitions we are interested can be split into two cases, depending on whether n is in a line by itself or not. If it is by itself, then there are $L(n-1, k-1)$ ways to pick the remaining lines. If n is not by itself then it has to be added to an already existing line. First we have to pick k lines from $n-1$ objects, then select which line we are inserting n into, then which place in that line it takes. There are $(n+k-1)L(n, k)$ ways of doing so.

These numbers are called the factorial numbers of the third kind due to the following formula which shows that they are coefficients of $x^{\overline{n}}$ in terms of $x^{\underline{k}}$.

Exercise 43: Use a counting argument to prove

$$x^{\overline{n}} = \sum_{k=0}^n L(n, k) x^{\underline{k}}.$$

Exercise 44: Using the previous exercise, prove that

$$x^{\overline{n}} = \sum_{k=0}^n (-1)^{n-k} L(n, k) x^{\overline{k}}.$$

Exercise 45: Use a counting argument to prove

$$L(n+1, k+1) = \sum_{i=k}^n L(i, k) (n+k+1)^{n-i}.$$

2.6 Sum of powers formulas

One digression which will still be beneficial is to investigate the sum of the n^{th} powers of the first k numbers

$$p_k(n) = \sum_{i=1}^k i^n$$

We want a nice closed form of this expression as polynomial in k of degree $n+1$, but if we were to write it in terms of x^i then we would need to go farther afield than what we have been able to accomplish so far. In order to use what we already know, we need to make use of the following formula

Exercise 46: Use a counting argument to prove

$$f_k(n) = \sum_{i=0}^k i^n = \frac{(k+1)^{n+1}}{n+1}.$$

Hint Define J_K^N to be the set of injections from N to K such that $f(i) < f(n)$ for $1 \leq i < n$. Then $|J_K^N| = f_{k-1}(n-1)$.

With this summation formula for factorials in mind and the previous exercise stating $x^n = \sum_{k=0}^n \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} x^k$ we can return to the sum of powers and rewrite it as

$$p_k(n) = \sum_{i=0}^k \sum_{j=0}^n \left\{ \begin{smallmatrix} n \\ j \end{smallmatrix} \right\} i^j,$$

but since these sums are over independent ranges we can reverse their order and factor out the Stirling coefficient of the second kind to get

$$p_k(n) = \sum_{j=0}^n \left\{ \begin{smallmatrix} n \\ j \end{smallmatrix} \right\} \sum_{i=0}^k i^j = \sum_{j=0}^n \left\{ \begin{smallmatrix} n \\ j \end{smallmatrix} \right\} \frac{(k+1)^{j+1}}{j+1}$$

This proves our original assertion that $p_k(n)$ is a polynomial in k with degree $n+1$.

Exercise 47: Use a counting argument to prove

$$p_k(n) = \sum_{j=0}^k \left\{ \begin{smallmatrix} n \\ j \end{smallmatrix} \right\} \frac{(k+1)^{j+1}}{j+1} = \sum_{j=0}^k \binom{k+1}{j+1} j! \left\{ \begin{smallmatrix} n \\ j \end{smallmatrix} \right\}.$$

Hint Define P_K^N to be the set of functions from N to K such that $f(i) < f(n)$ for $1 \leq i < n$. Then $|P_K^N| = p_{k-1}(n-1)$.

We could generate an inductive procedure to find these polynomials written in terms of n^k using the following identity of Pascal

Exercise 48: Prove the following

$$(k+1)^{n+1} - 1 = \sum_{i=0}^n \binom{n+1}{i} p_k(i).$$

Hint use $(j+1)^{k+1} - j^{k+1}$ to create a telescoping series.

2.7 Inversions and q-Binomials

3 Grids and words

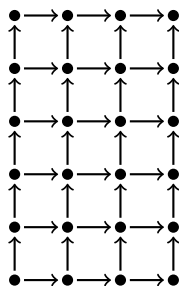
It may seem like we made a huge detour into topics barely related to digraphs and mazes since we discussed maze composition, but all of that was setup so that we can appreciate the beauty of these proofs.

3.1 Grid(n,k)

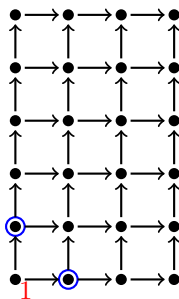
The star of the show for this section is the maze $Grid(n, k)$. The vertices of this maze and the edges between them can be simply described in a coordinate plane. For our maze $grid(n, k)$ we define V to be the set of points

$$\{P_{i,j} = (i, j) \mid 0 \leq i \leq k, 0 \leq j \leq n - k\}$$

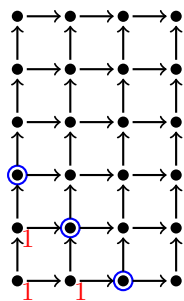
and the edge set to be all edges from $A_{i,j}$ to $A_{i+1,j}$ and from $A_{i,j}$ to $A_{i,j+1}$, when both points exist. That means only $A_{0,0}$ can be our start and only $A_{k,n-k}$ can be our end. Here is $Grid(8, 3)$ for example



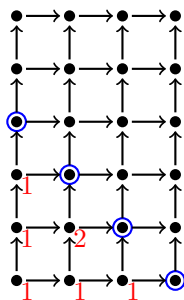
This maze looks pretty unremarkable until you perform the water pipes algorithm on it. In the first step we simply label the start with a 1 and circle the next 2 vertices that need to be weighed.



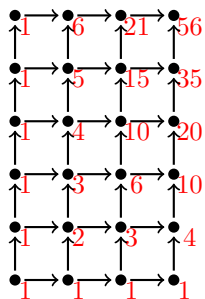
The next step is not that impressive either;



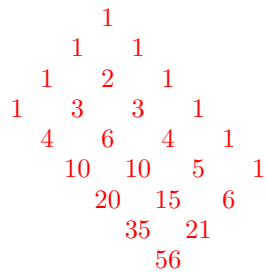
but if you are perceptive you might notice something in the third step.



In order to make the pattern clearer before we state the theorem, we will present the maze after the algorithm has completed



If we just forget the maze and rotate the position of the numbers we see



what appears to be a portion of the binomial triangle. Succinctly, we have observed that

$$|Grid(8, 3)| = 56 = \binom{8}{3}.$$

It is reasonable to expect that

Theorem 1 the number of solutions to $Grid(n, k)$ is $\binom{n}{k}$.

Proof Observe that $|Grid(n, 0)| = |Grid(n, n)| = 1$, and by writing $n = a + b$, $k = a$ then we have

$$|Grid(a + b, a)| = |Grid(a + b - 1, a - 1)| + |Grid(a + b - 1, a)|$$

which is exactly the binomial recurrence. Since the 2-parameter sequences (counting triangles) have the same initial conditions, and the same recurrence relation, they are equal.

This is not the most inspired proof that our maze contains a portion of the binomial triangle, but we will run into a better one when we discuss words. For now we are going to return to the Vandermonde Convolution and the Hockey stick lemma, showing that grid mazes make their proofs straightforward.

Theorem 2 (Vandermonde Convolution): Given nonnegative integers m, n, k we have

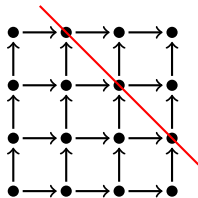
$$\binom{m + n}{k} = \sum_{i=0}^k \binom{m}{i} \binom{n}{k - i}$$

Proof Consider the maze $Grid(m + n, k)$, and the set of points

$$l_m = \{ (i, j) \in V(Grid(m + n, k)) \mid i + j = m \}.$$

We will divide solutions of $Grid(m + n, k)$ into subsets according to which of the points (i, j) they pass through, which will let us do casework. For case i we are looking at the paths which start at $(0, 0)$ and go through $(m - i, i)$ before they reach the point $(m + n - k, k)$. We can view such a solution in the restricted maze $Grid(m, i) \times Grid(n, k - i)$ which we know has $\binom{m}{i} \binom{n}{k - i}$ solutions from our previous theorem and our discussion of maze composition. By summing over all i , we count the total number of solutions of $grid(m + n, k)$, and achieve the result.

Lets look at an example of this when $m = 4$, $n = 2$, and $k = 3$. Then our maze is $Grid(6, 3)$ which we know has 20 solutions, and with the line l_4 it looks like looks like



The number of solutions which pass through the point $(3, 1)$ is $\binom{4}{3} \binom{2}{0} = 4$, the number of solutions through $(2, 2)$ is $\binom{4}{2} \binom{2}{1} = 12$, and the number of solutions through $(1, 3)$ is $\binom{4}{3} \binom{2}{0} = 4$, which we can confirm sum up to 20.

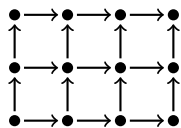
These maze proofs are quite powerful, and we will see variants of them again in new contexts, but the rest of this section is devoted to applications grid mazes specifically. As promised we will prove the Hockey stick identity next.

Theorem 3 (hockey stick identity): For all nonnegative n, k we have

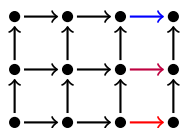
$$\binom{n+1}{k+1} = \sum_{i=k}^n \binom{i}{k}.$$

Proof The left side suggests we should consider the maze $Grid(n+1, k+1)$, the right side suggests we should investigate possible casework involving the mazes $Grid(i, k)$. The trick is to consider where the last right step occurs. For case i , we will count the amount of solutions that have the point $(k, i-k)$ as the tail of their final right step. There are $\binom{i}{k}$ paths through the maze to the point $(k, i-k)$, and then after we make our final right step to $(k+1, i-k)$, we can only perform up steps until we reach the end. Summing over all i completes the proof.

Lets look at another example so that the details of the proof can benefit from visual clarity. Suppose that $n = 4$ and $k = 2$. Then we know the number of solutions to the maze $Grid(5, 3) = 10$, and it looks like



Lets color the possible last right steps red, purple, and blue respectively



Since we can only move up after taking a colored arrow, we know that the number of paths through the red arrow is $\binom{2}{2} = 1$, through the purple arrow is $\binom{3}{2} = 3$, and through the blue arrow is $\binom{4}{2} = 6$ which all add up to 10.

Exercise 49 (mirror hockey stick identity): Use a maze to prove

$$\binom{n+1}{n-k} = \sum_{i=0}^{n-k} \binom{k+i}{i}.$$

Exercise 50: We will be discussing mazes in the context of factorial coefficient identities in a later section. What should a maze look like if it is going to model properties of factorial coefficients? Try to draw some factorial mazes.

This exercise suggests that we need to consider mazes like $Grid(n, k)$, but maybe with restrictions on which vertices or edges we include or exclude.

3.2 Words and anagrams

In this subsection we will be connecting mazes to words, and prepare more ideas for future investigations. For our purposes, an **alphabet** A is a finite set of symbols. A **word** from our alphabet is a list of symbols from A . For words we typically just write the symbols concatenated in the order they appear in the list. So for instance if $A = \{a, b, c\}$, then a possible word would be $[a, c, b, c, a]$ but we would write it as $acbca$.

Given an alphabet $A = \{a_1, a_2, \dots, a_n\}$ and a list of numbers $B = [b_1, b_2, \dots, b_n]$, a question we like to ask is how many words are there that have b_1 a_1 's, b_2 a_2 's, $\dots$, and b_n a_n 's. If two words have the same amount of each of the symbols, then we say that they are **anagrams** of each other. Using our previous example, $ccaab$ is an anagram of $acbca$, but how many anagrams could we find for these words?

Since anagrams are defined up to the number of times each symbol appears in the word, a convenient way to proceed is to combine our set A and list B into a letter multiset $\mathcal{A}$ with elements from A where the multiplicity of the element a_i is b_i , or more explicitly

$$\mathcal{A} = [b_1 \cdot a_1, b_2 \cdot a_2, \dots, b_n \cdot a_n].$$

Then we define the set $Anagram(\mathcal{A})$ to be the set of words from the alphabet A subject to the constraint that the words have b_i instances of the symbol a_i .

Exercise 51: Use a counting argument to prove that

$$|Anagram(\mathcal{A})| = \frac{(b_1 + b_2 + \dots + b_n)!}{b_1! b_2! \dots b_n!}$$

Hint First choose the b_1 spots out of $b_1 + b_2 + \dots + b_n$ possible locations for the a_1 s, then from the remaining $b_2 + \dots + b_n$ spots, choose where the b_2 spots for the a_2 s, and so on.

If we focus on the case of a letter set over an alphabet with two letters $\mathcal{G} = [k \cdot r, (n - k) \cdot u]$, we can find a bijection between paths in $Grid(n, k)$ with elements of $Anagram(\mathcal{G})$. A solution to $Grid(n, k)$ is a sequence of k right moves and $n - k$ up moves, so we can construct a word that has its i^{th} letter from the left be an r if that move is a right step, and a u if it is an up step.

Exercise 52: Define the multinomial coefficient to be the symbol

$$\binom{n_1 + n_2 + \dots + n_k}{n_1, n_2, \dots, n_k} = \frac{(n_1 + n_2 + \dots + n_k)!}{n_1! n_2! \dots n_k!}$$

subject to the convention that $\binom{0}{0, 0, \dots, 0} = 1$ for any number of 0s. Prove that this symbol satisfies the recurrence

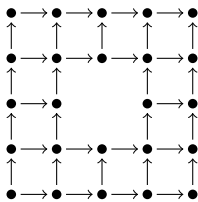
$$\binom{n_1 + n_2 + \dots + n_k}{n_1, n_2, \dots, n_k} = \sum_{i=1}^k \binom{n_1 + n_2 + \dots + n_k - 1}{n_1, n_2, \dots, n_i - 1, \dots, n_k}.$$

What should a maze look like which models properties of multinomial coefficients? Try to draw one.

Exercise 53: Suppose we know the size of our letter set $\mathcal{X}$ is n , and that we have up to k possible different types of symbols in our letter set. How many different possibilities for our letter set are there? How many are there if we know there is at least one of all k types of symbols?

3.3 Grids with forbidden positions

Back in section 1 the very first maze we introduced was



Which is almost $Grid(8, 4)$ except we are not allowed to visit the point $(2, 2)$ on our way to $(4, 4)$. Of course water pipes could give us the number of solutions, but we can also do less arithmetic if we simply subtracted the number of solutions to $Grid(8, 4)$ passing through our forbidden point. A path passing through this forbidden point can be seen as a solution to the maze $Grid(4, 2) \times Grid(4, 2)$ which has $6^2 = 36$ solutions. Since we have 70 solutions to $Grid(8, 4)$, there are 34 solutions to our restricted maze.

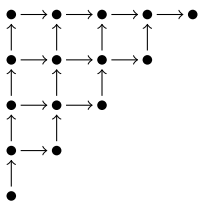
This method is not always preferable over water pipes. The amount of arithmetic we have to do with inclusion/exclusion goes up while the amount of arithmetic we have to do for water pipes goes down as we take away more points from our grid maze. It is up to the reader to decide which method is better for them in the context they are working.

Exercise 54: How many solutions to $Grid(20, 7)$ are there if we are not allowed to visit the points $(1, 4)$ or $(5, 11)$? Compare both of our methods for solving this problem.

3.4 Dyck paths and Catalan numbers

Certain restrictions on grid mazes produce even more interesting counting sequences.

Consider the maze $Grid(2n, n)$ where we are not allowed to visit (i, j) if $i > j$, or put another way, we are allowed to touch, but not cross, the line $y = x$. We call this maze $Cat(n)$ and say it has C_n solutions. The numbers C_n are widely known as the Catalan numbers after mathematician Eugene Charl  s Catalan, but they were first studied by Euler and Minggatu. Below is the case when $n = 4$.



A solution to this maze is commonly called a **Dyck path**. Of course we could use water pipes to solve this maze, but really we would like a formula in terms of n . However, this time we will break our pattern and find a nonlinear recurrence relation for our numbers before we obtain our formula. Later we will find a second recurrence using another type of visualization of what these numbers are counting.

Lemma 5: the Catalan numbers C_n satisfy the following recurrence:

$$C_{n+1} = \sum_{i=0}^n C_i C_{n-i}$$

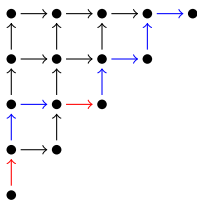
where we define $C_0 = 1$ by agreeing that there is only one way to solve a maze whose start is the same as its end.

Proof We can divide the paths into cases, where a path is counted by case i if the first time the path touches the line $y = x$ (after leaving the start) is at the point (i, i) . We can split a path in case i into two smaller paths. The first part of the path is an up arrow, followed by a solution to $Cat(i - 1)$, and then a right arrow, which guarantees that this part of the path does not touch $y = x$ until (i, i) . The second part of the path is a solution to $Cat(n + 1 - i)$. By our discussion of maze composition we know there are $1 \times C_{i-1} \times 1 \times C_{n+1-i} = C_{i-1} \times C_{n+1-i}$ possible mazes in each case, so by summing from $i = 1$ to $i = n + 1$ we get

$$C_{n+1} = \sum_{i=1}^{n+1} C_{i-1} C_{n+1-i}$$

and achieve the result by re indexing.

Here is an example of a path in $Cat(4)$ which we have split up according to our instructions



This recurrence relation is useful for proving other types of mathematical objects are counted by C_n (these numbers pop up over and over again in counting) but it is pretty unwieldy as far using it to compute actual values goes. Fortunately there is an explicit formula that we can use instead.

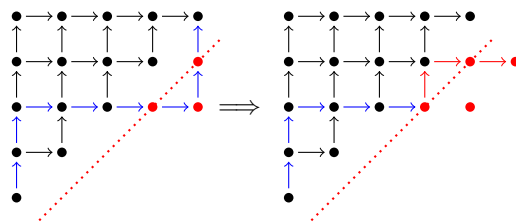
Theorem 4: The Catalan number C_n can be computed as

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

Proof We will proceed by identifying $Cat(n)$ with $Grid(2n, n)$ subject to restrictions, then subtract the paths in $Grid(2n, n)$ which violate our restrictions from the total amount of paths. A path which first crosses the line $y = x$ at (i, i) can be split into a path from $(0, 0)$ to $(i + 1, i)$, and a path from $(i + 1, i)$ to (n, n) . If we take this second path and reflect it across the diagonal line $y = x - 1$, then we get a new path from $(i + 1, i)$ to $(n + 1, n - 1)$. In this way every path which violates our restrictions in $Cat(n)$ can be bijectively assigned to a path in $Grid(2n, n + 1)$. Therefore the total number of paths in $Cat(n)$ is

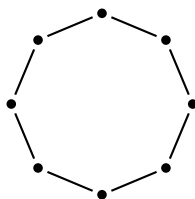
$$C_n = \binom{2n}{n} - \binom{2n}{n+1} = \binom{2n}{n} - \frac{n}{n+1} \binom{2n}{n} = \frac{1}{n+1} \binom{2n}{n}.$$

To help the reader visualize this proof we have illustrated an example of the transformation sending an invalid path in $Cat(n)$ to a valid path in $Grid(2n, n + 1)$

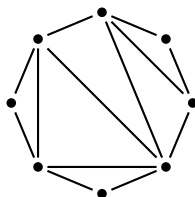


This proof technique originated from French mathematician Désiré André.

As stated before, the numbers C_n count a wide variety of distinct mathematical objects. In this section we will see an example from a familiar context, and next section we will see another example. Back in section 1, in an exercise, we established Euler's formula relating the number of faces, edges, and vertices of a planar graph by considering triangulations of that graph. In a planar graph, edges are not allowed to cross each other when we draw the graph, so we impose the same restriction on our triangulations. For this subsection we will focus on graphs that are cycles and drawn as convex n -gons. For example when $n = 8$ we have



and a possible triangulation of this graph is given by



Given a convex n -gon, how many ways can we triangulate it with non crossing segments? Since we introduced this question in the section about Catalan numbers you could probably guess that they would be involved.

Exercise 55: Prove that for $n \geq 0$ the number of triangulations of a convex $(n + 2)$ -gon is given by C_n , where we agree that there is one triangulation of a segment. **Hint** Start with a triangle and glue a convex k -gon to one side, and a convex $(n - k)$ -gon to the other side.

Exercise 56: Suppose that concert tickets are 5 dollars, and we have n people with a single 5 dollar bill, and n people with a 10 dollar bill. If you start with no cash, how many ways can the people line up so that you can always break a 10 dollar bill if you need to?

Exercise 57: Suppose we have the letter multiset $P_n = [n \cdot (\ n \cdot)]$, how many anagrams can we find that always have more ('s than)'s when read from left to right? An example and nonexample when $n = 3$ are $((()())$ and $)()((()$.

3.5 Galton Mazes

Our discussion of mazes can also help us prepare for discussions about probability if we allow the mazes to have possibly many starts and ends.

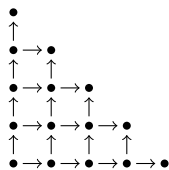
For this subsection we will consider connected digraphs that do not contain any cycles which have one vertex with indegree and at least two vertices with outdegree zero. In such a maze we say that the vertex with indegree zero is the start, and the vertices with outdegree zero are the ends. As before, a solution to the maze is a path from the start to one of the ends.

A possible question we could investigate is given an end v , what is the probability that a solution of our maze ends at v ? It turns out that this is not a hard problem to answer with water pipes, just divide the number of solutions to v by the total number of solutions of the maze.

We will define a new maze $Galton(n)$ to be a maze starting at $(0, 0)$ where we are allowed to step one to the right or one up until we reach a point on the line $x + y = n$, or more formally

$$Galton(n) = \bigcup_{i=0}^n Grid(n, i).$$

An example when $n = 4$ is



The name comes from the popular demonstration of statistical concepts using marbles.

Exercise 58: Use $Galton(n)$ to prove

$$2^n = \sum_{k=0}^n \binom{n}{k}.$$

Hint: A solution of $Galton(n)$ corresponds to a word of n letters from two types.

Exercise 59: Suppose we are constructing a random solution to $Galton(n)$ by deciding if the i^{th} step of n is a right step with probability p , and an up step otherwise. What is the probability we end at the point $(i, n - i)$?

Exercise 60: Come up with a maze which proves

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.$$

Exercise 61: Come up with a maze which proves

$$(a + b + c)^n = \sum_{i+j+k=n} \binom{n}{i, j, k} a^i b^j c^k$$

where i, j, k are nonnegative integers.

3.6 Simple factorial mazes

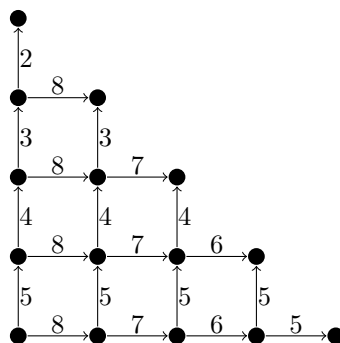
The exercises of the previous subsection suggested that there is a simple maze interpretation of the expression k^n which we could use to prove

$$k^n = \sum_{a_1 + a_2 + \dots + a_k = n} \binom{n}{a_1, a_2, \dots, a_k}.$$

In this subsection we will come up with a maze which proves

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.$$

It is a modification of a maze we have seen before, but now we will explicitly weight the edges of the maze so that the weight of a path is the product of the weights of all the edges. For falling factorials we will have two dimensions labeled with descending terms of a and b minus a nonnegative integer. For instance when $a = 8$ and $b = 5$ with $n = 4$ is



The general construction $(a + b)^n$ is almost the same as the galton maze but the edges from (i, j) to $(i + 1, j)$ are labeled with $a - i$ and the edges from (i, j) to $(i, j + 1)$ are labeled with $b - j$.

Now suppose we wish to count the solutions to our maze. To establish the left side, we need to understand what happens to the number of options we have after each step. We start out with $a + b$ possibilities for which path we take. After taking a step, one option is removed so we have $a + b - 1$. After two steps we have two options removed so now we have $a + b - 2$ possibilities. After n steps we have $a + b - n + 1$ options left to us. Multiplying over every choice gives us the desired expression. To establish the right side, suppose we count the paths by casework on which end we terminate at. If we end at $(k, n - k)$, then there are $\binom{n}{k}$ ways to choose which vertices we visit along the way. For each selection of vertices we get $a^k b^{n-k}$ possibilities for the paths we could take through those vertices. Summing over all cases k from 0 to n establishes the identity.

Exercise 62 Define $a^{n:t} = a(a - t)(a - 2t) \dots (a - (n - 1)t)$. Come up with mazes which prove

$$(a + b + c)^n = \sum_{i+j+k=n} \binom{n}{i, j, k} a^i b^j c^k, \quad (a + b + c)^{\bar{n}} = \sum_{i+j+k=n} \binom{n}{i, j, k} a^{\bar{i}} b^{\bar{j}} c^{\bar{k}},$$

$$(a + b)^{n:t} = \sum_{k=0}^n \binom{n}{k} a^{k:t} b^{(n-k):t}.$$

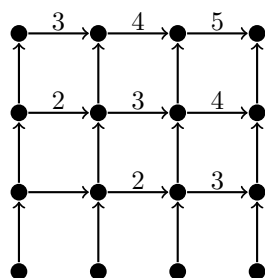
3.7 Stirling Mazes

So far we have successfully used mazes to prove a lot of properties of binomial symbols and factorials in a straightforward way. These ideas can also be applied to our factorial coefficients to sidestep the need for creative counting arguments. Recall that the factorial coefficients of the first kind satisfy the recurrence relation

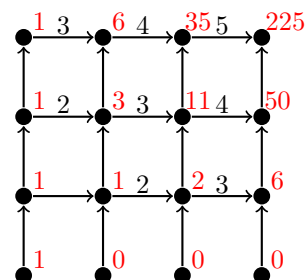
$$\begin{bmatrix} n \\ k \end{bmatrix} = \begin{bmatrix} n-1 \\ k-1 \end{bmatrix} + (n-1) \begin{bmatrix} n-1 \\ k \end{bmatrix}$$

which we can use to modify the grid maze. Define the maze $S1(n, k)$ with the same vertex set as $Grid(n, k)$ and an edge set which has an edge from (i, j) to $(i, j+1)$ for $0 \leq i \leq k$ and $0 \leq j < n-k$ and $i+j$ edges from (i, j) to $(i+1, j)$ for $1 \leq i < k$ and $0 \leq j \leq n-k$.

For example we have $S1(6, 3)$



If we declare that the bottom left point is our start and all other points with indegree 0 are labeled with 0 then the water pipes algorithm gives us



Exercise 63 Show that $|S1(n, k)| = |S1(n-1, k-1)| + (n-1)|S1(n-1, k)|$. Use this recurrence to conclude $|S1(n, k)| = \begin{bmatrix} n \\ k \end{bmatrix}$.

Exercise 64: Use a maze argument to prove

$$\begin{bmatrix} n+1 \\ k+1 \end{bmatrix} = \sum_{i=k}^n \begin{bmatrix} n \\ i \end{bmatrix} n^{n-i}.$$

Exercise 65: Use a maze argument to prove

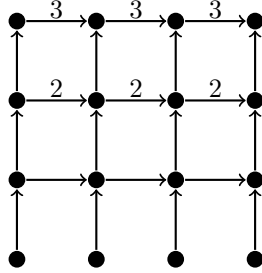
$$\begin{bmatrix} n+k+1 \\ k \end{bmatrix} = \sum_{i=0}^k (n+i) \begin{bmatrix} n+i \\ i \end{bmatrix}$$

Recall that the factorial coefficients of the second kind satisfy the recurrence relation

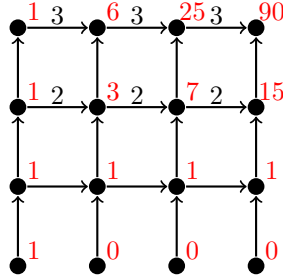
$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \left\{ \begin{matrix} n-1 \\ k-1 \end{matrix} \right\} + k \left\{ \begin{matrix} n-1 \\ k \end{matrix} \right\}$$

which we can use to modify the grid maze. Define the maze $S2(n, k)$ with the same vertex set as $Grid(n, k)$ and an edge set which has an edge from (i, j) to $(i, j+1)$ for $0 \leq i \leq k$ and $0 \leq j < n-k$ and j edges from (i, j) to $(i+1, j)$ for $1 \leq i < k$ and $0 \leq j \leq n-k$.

For example we have $S2(6, 3)$



If we declare that the bottom left point is our start and all other points with indegree 0 are labeled with 0 then the water pipes algorithm gives us



Exercise 63 Show that $|S2(n, k)| = |S2(n-1, k-1)| + k|S2(n-1, k)|$. Use this recurrence to conclude $|S2(n, k)| = \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$.

Exercise 33: Use a maze argument to prove

$$\left\{ \begin{matrix} n+1 \\ k+1 \end{matrix} \right\} = \sum_{j=k}^n (k+1)^{n-j} \left\{ \begin{matrix} j \\ k \end{matrix} \right\}.$$

Exercise 34: Use a maze argument to prove

$$\left\{ \begin{matrix} n+k+1 \\ k \end{matrix} \right\} = \sum_{j=0}^k j \left\{ \begin{matrix} n+j \\ k \end{matrix} \right\}.$$

- 4 Linear Algebra With Digraphs
- 5 Counting Symbols as change of basis coefficients
- 6 Algebraic Derivatives

7 Appendix A Data structures

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8 Appendix B Counting principles

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9 Appendix C Simple sequences and series

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10 Solutions

Solution 1: Every edge contributes $+1$ to the degree of 2 vertices.